

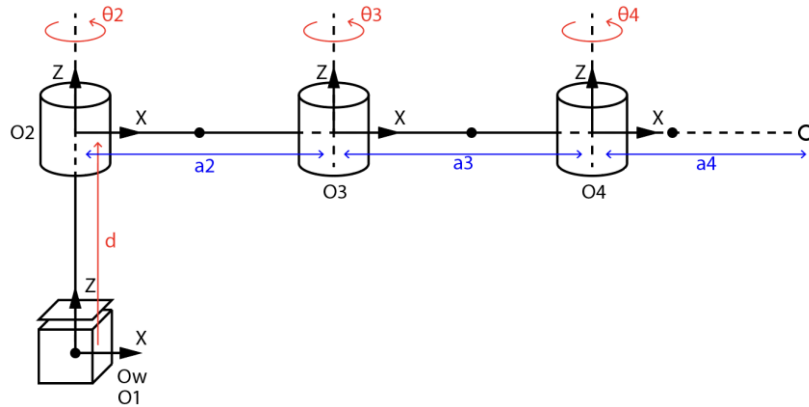
I Modelling the robot

For WTR robots, consider 3 connected links on the plane, with for following parameters:

	Link 1	Link 2	Link 3	Link 4
Mass	m_1	m_2	m_3	m_4
Length	$L_1 = 0$	L_2	L_3	L_4
Joint	d_1	θ_2	θ_3	θ_4
Center of mass distance (w.r.t. head)	$L_{c1} = 0$	L_{c2}	L_{c3}	L_{c4}
Center of mass (w.r.t. head)	c_1	c_2	c_3	c_4
Center of mass (w.r.t. world/base)	P_{c1}	P_{c2}	P_{c3}	P_{c4}
Inertia (w.r.t. CoM)	N/A	I_{zz2}	I_{zz3}	I_{zz4}

Green parameters are provided during design and considered as constants, blue parameters are derived parameters and are dependent of variables. Red parameters are the variables.

Associated frames and configuration are shown below:



The corresponding modified DH parameters are:

i-th Joint	θ_i	a_{i-1}	d_i	α_{i-1}	Type
1	0	0	$q_1 = d_1$	0	Prismatic
2	$q_2 = \theta_2$	L_1	0	0	Revolute
3	$q_3 = \theta_3$	L_2	0	0	Revolute
4	$q_4 = \theta_4$	L_3	0	0	Revolute
5	N/A	L_4	N/A	0	N/A

II Deriving kinematic matrices

1. Using DH matrices

According to the modified DH convention, the transformation of a state in homogeneous coordinate is done by modelling the parameters θ, d, a, α sequentially:

$$T_i^{i-1} = Rot_X(\alpha_{i-1})Trans_X(a_{i-1})Rot_Z(\theta_i)Trans_Z(d_i)$$

$$= \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \alpha_{i-1} \cos \theta_i & -\sin \alpha_{i-1} & -\sin \alpha_{i-1} d_i \\ \sin \theta_i \sin \alpha_{i-1} & \sin \alpha_{i-1} \cos \theta_i & \cos \alpha_{i-1} & \cos \alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note the 0-th DH parameter is considered to be zero.

Since for WTR/SCARA-type robots, we always have $\alpha_i = 0$, T_i^{i-1} can be simplified:

$$T_i^{i-1} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For all 4 joints, we have the total transformation matrix:

$$T_4^0 = T_1^0 T_2^1 T_3^2 T_4^3$$

Or generally for the i-th link, the transformation matrix can be defined as:

$$T_i^0 = \prod_{j=1}^i T_j^{j-1}$$

And for mass center c_i defined in the local frame of each link, they can be transformed into world frame:

$$c_i^0 = T_i^0 c_i = P_{ci}$$

Deriving the mass center of each link is crucial for constructing the Lagrangian – we need the kinetic energy and potential energy for each link, which are usually defined around the link's mass center.

2. Using direct geometrical calculation

Another way of deriving the mass center location is through the geometry of the robot, which involves no transformation matrices. This method can be intuitive for robots with only a few joints.

First, we derive position of the end point of each link:

$$\begin{aligned} p_0 &= (0,0,0)^T \\ p_1 &= p_0 + (0,0,d_1)^T \\ &= (0,0,d_1)^T \\ p_2 &= p_1 + (L_2 \cos \theta_2, L_2 \sin \theta_2, 0)^T \\ &= (L_2 \cos \theta_2, L_2 \sin \theta_2, d_1)^T \\ p_3 &= p_2 + (L_3 \cos(\theta_2 + \theta_3), L_3 \sin(\theta_2 + \theta_3), 0)^T \\ &= (L_2 \cos \theta_2 + L_3 \cos(\theta_2 + \theta_3), L_2 \sin \theta_2 + L_3 \sin(\theta_2 + \theta_3), d_1)^T \\ p_4 &= p_3 + (L_4 \cos(\theta_2 + \theta_3 + \theta_4), L_4 \sin(\theta_2 + \theta_3 + \theta_4), 0)^T \\ &= \begin{pmatrix} L_2 \cos \theta_2 + L_3 \cos(\theta_2 + \theta_3) + L_4 \cos(\theta_2 + \theta_3 + \theta_4) \\ L_2 \sin \theta_2 + L_3 \sin(\theta_2 + \theta_3) + L_4 \sin(\theta_2 + \theta_3 + \theta_4) \\ d_1 \end{pmatrix} \end{aligned}$$

We can derive the center of mass w.r.t. the rotation axis (origin) of each link.

$$\begin{aligned} c_1 &= (0,0,d_1)^T \\ c_2 &= (L_{c2} \cos \theta_2, L_{c2} \sin \theta_2, 0)^T \\ c_3 &= (L_{c3} \cos(\theta_2 + \theta_3), L_{c3} \sin(\theta_2 + \theta_3), 0)^T \\ c_4 &= (L_{c4} \cos(\theta_2 + \theta_3 + \theta_4), L_{c4} \sin(\theta_2 + \theta_3 + \theta_4), 0)^T \end{aligned}$$

Combined we have the position of each mass center in the world frame:

$$\begin{aligned}
P_{c1} &= c_1 + p_0 = (0, 0, d_1)^T \\
P_{c2} &= c_2 + p_1 = (L_{c2} \cos \theta_2, L_{c2} \sin \theta_2, d_1)^T \\
P_{c3} &= c_3 + p_2 = (L_{c3} \cos(\theta_2 + \theta_3) + L_2 \cos \theta_2, L_{c3} \sin(\theta_2 + \theta_3) + L_2 \sin \theta_2, d_1)^T \\
P_{c4} &= c_4 + p_3 \\
&= \begin{pmatrix} L_{c4} \cos(\theta_2 + \theta_3 + \theta_4) + L_2 \cos \theta_2 + L_3 \cos(\theta_2 + \theta_3) \\ L_{c4} \sin(\theta_2 + \theta_3 + \theta_4) + L_2 \sin \theta_2 + L_3 \sin(\theta_2 + \theta_3) \\ d_1 \end{pmatrix}
\end{aligned}$$

With the help of the CAS system in MATLAB we can verify the two methods yield the same result.

III Forming the Lagrangian

The Lagrangian is defined by $L = K - U$, in which K is the total kinetic energy of the system and U is the total potential energy of the system. Physically the above energies can be defined using the object's mass center:

$$K_i = \frac{1}{2} m_i v_{ci}^T v_{ci} + \frac{1}{2} \omega_i^T I_i \omega_i$$

For WTR/SCARA-type robots, all joints rotate around their Z axis, therefore the second term can be simplified to only involving the joint velocity and the inertia around the Z axis at the mass center:

$$K_i = \frac{1}{2} m_i v_{ci}^T v_{ci} + \frac{1}{2} I_{zzi} \dot{\theta}_i^2$$

1. Kinetic energy using joint velocity

Since in the end we want the relationship between joint torque and joint position, velocity, acceleration, etc., it's better to formulate kinetic energy in terms of the joint variables $q, \dot{q}, \ddot{q}$. This can be done by taking the derivative of mass centers w.r.t. joint variables, more specifically, joint velocities:

$$\begin{aligned}
K_i &= \frac{1}{2} \dot{q}^T D_i(q) \dot{q} \\
D_i &= \frac{\partial P_{ci}}{\partial \dot{q}}
\end{aligned}$$

$D_i(q)$ is called the inertia matrix of this link. The total kinetic energy K is formed by summing individual kinetic energy of each link:

$$K = \sum_{i=1}^n K_i = \frac{1}{2} \dot{q}^T M(q) \dot{q}$$

$M(q)$ is called the mass matrix of the robot. As later we'll see, this is one of the terms we want for calculating the inverse dynamics.

Deriving the mass matrix by taking derivatives can be a mess if you do it by hand. Instead, it can be done by utilizing link Jacobian:

$$J_i = [J_{vi}^T \ J_{\omega i}^T]^T$$

The link Jacobian transforms the joint velocity to velocity of the link's mass center in world frame.

Link Jacobian can be derived as:

$$J_i = \begin{bmatrix} r_{vi_1} & r_{vi_2} & \cdots & r_{vi_n} \\ r_{\omega i_1} & r_{\omega i_2} & \cdots & r_{\omega i_n} \end{bmatrix}_{6 \times n}$$

For revolute joints:

$$r_{vi_j} = \begin{cases} z_j \times (P_{ci} - p_j), & j \leq i \\ 0, & j > i \end{cases}$$

$$r_{\omega i_j} = \begin{cases} z_j, & j \leq i \\ 0, & j > i \end{cases}$$

For prismatic joints:

$$r_{vi_j} = \begin{cases} z_j, & j \leq i \\ 0, & j > i \end{cases}$$

$$r_{\omega i_j} = 0$$

z_j is the representation of Z axis of the j-th link in world frame, it's also the third column of the rotation matrix part in the total homogeneous matrix up till this link: p_j represents the origin of the j-th link, it's also the translational part of the total homogeneous matrix up till this link.

$$T_j^0 = \begin{bmatrix} R_j^0 & p_j \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} [x_j & y_j & z_j] & p_j \\ 0 & 1 \end{bmatrix}$$

In total the link Jacobian provides the overall contribution of each joint to the i-th mass center in world frame. Utilizing the link Jacobian, link velocity and angular velocity is just:

$$v_{ci} = J_{vi} \dot{q}$$

$$\omega_i = J_{\omega i} \dot{q}$$

In order to calculate energy, we need the inertia of each link to be represented in the world frame. This can be done by utilizing the rotation part of the total homogeneous matrix up till this link:

$$I_i^0 = R_i^0 I_i R_i^{0T}$$

Finally, we can rewrite the kinetic energy in these terms:

$$\begin{aligned} K_i &= \frac{1}{2} m_i (J_{vi} \dot{q})^T (J_{vi} \dot{q}) + \frac{1}{2} (J_{\omega i} \dot{q})^T I_i^0 (J_{\omega i} \dot{q}) \\ &= \frac{1}{2} \dot{q}^T m_i J_{vi}^T J_{vi} \dot{q} + \frac{1}{2} \dot{q}^T J_{\omega i}^T I_i^0 J_{\omega i} \dot{q} \\ &= \frac{1}{2} \dot{q}^T D_i(q) \dot{q} \end{aligned}$$

Among which:

$$D_i(q) = m_i J_{vi}^T J_{vi} + J_{\omega i}^T I_i^0 J_{\omega i}$$

And adding the above kinetic energy for all links we have the explicit form of total kinetic energy, also the mass matrix.

For the potential energy, it's only related to gravity (no springs are involved):

$$U_i = [0 \quad 0 \quad 1] m_i g p_i$$

2. Kinetic energy using Cartesian velocity

If you already derived the mass center location in analytical form, then the most intuitive way is to get velocity simply by taking its derivative w.r.t. time.

For later simplicity we calculate local direction of each link in world frame:

$$r_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad r_2 = \begin{pmatrix} -\sin \theta_2 \\ \cos \theta_2 \\ 0 \end{pmatrix} \quad r_3 = \begin{pmatrix} -\sin(\theta_2 + \theta_3) \\ \cos(\theta_2 + \theta_3) \\ 0 \end{pmatrix} \quad r_4 = \begin{pmatrix} -\sin(\theta_2 + \theta_3 + \theta_4) \\ \cos(\theta_2 + \theta_3 + \theta_4) \\ 0 \end{pmatrix}$$

Mass center velocity can be seen as a combination of the above directions:

$$\begin{aligned} v_{c1} &= \dot{d}_1 r_1 \\ v_{c2} &= L_{c2} \dot{\theta}_2 r_2 + \dot{d}_1 r_1 \\ v_{c3} &= L_{c3} (\dot{\theta}_2 + \dot{\theta}_3) r_3 + L_2 \dot{\theta}_2 r_2 + \dot{d}_1 r_1 \\ v_{c4} &= L_{c4} (\dot{\theta}_2 + \dot{\theta}_3 + \dot{\theta}_4) r_4 + L_3 (\dot{\theta}_2 + \dot{\theta}_3) r_3 + L_2 \dot{\theta}_2 r_2 + \dot{d}_1 r_1 \end{aligned}$$

We then derive kinetic and potential energy for each link.

Kinetic energy (described per link as the combination of rotational and translational movement):

$$\begin{aligned} K_1 &= \frac{1}{2} m_1 v_{c1}^2 \\ K_2 &= \frac{1}{2} I_{zz2} \dot{\theta}_2^2 + \frac{1}{2} m_2 v_{c2}^2 \\ K_3 &= \frac{1}{2} I_{zz3} (\dot{\theta}_2 + \dot{\theta}_3)^2 + \frac{1}{2} m_3 v_{c3}^2 \\ K_4 &= \frac{1}{2} I_{zz4} (\dot{\theta}_2 + \dot{\theta}_3 + \dot{\theta}_4)^2 + \frac{1}{2} m_4 v_{c4}^2 \end{aligned}$$

Potential energy:

$$\begin{aligned} U_1 &= m_1 g d_1 \\ U_2 &= m_2 g d_1 \\ U_3 &= m_3 g d_1 \\ U_4 &= m_4 g d_1 \end{aligned}$$

IV Solving the Euler-Lagrange equation

Given Lagrangian L , we have total generalized (non-conservative) force defined by the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = F$$

Inverse dynamics involves solving this equation for F .

Since L is a function of $q, \dot{q}$ only, $\frac{\partial L}{\partial \dot{q}}$ is also a function of $q, \dot{q}$ only. This means the time derivative term can be expanded using chain rule:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial^2 L}{\partial \dot{q} \partial q} \dot{q} + \frac{\partial^2 L}{\partial \dot{q}^2} \ddot{q}$$

Which yields:

$$F = \frac{\partial^2 L}{\partial \dot{q} \partial q} \dot{q} + \frac{\partial^2 L}{\partial \dot{q}^2} \ddot{q} - \frac{\partial L}{\partial q}$$

Since we have the Lagrangian in analytical form, we can calculate each of the derivatives effectively and accurately using the technique of reverse mode automatic differentiation.

Actually, the Hessian of Lagrangian w.r.t. general velocity $\dot{q}$ is equivalent to the mass matrix $M(q)$, and one can

directly notice that the above equation is the exact motion equation:

$$F = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q)$$

Where:

$$M(q) = \frac{\partial^2 L}{\partial \dot{q}^2}$$
$$C(q, \dot{q}) = \frac{\partial^2 L}{\partial \dot{q} \partial q}$$
$$g(q) = -\frac{\partial L}{\partial q}$$

Note that in this direct method, the Coriolis and centrifugal matrix $C(q, \dot{q})$ and gravity term $g(q)$ is not unique, and it can be different from the same equation derived from other methods.